

LANGEBAAWEG, South Africa

WMO: **687140**Lat: **32.9725S**Lon: **18.1578E**Elev: **102**StdP: **14.64**Time Zone: **2.00 (E02)**Period: **96-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.4	41.2	30.9	25.2	68.1	33.5	28.1	60.6	24.6	59.6	22.4	59.7	2.8	190	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.0	94.9	66.6	91.1	65.7	87.4	64.6	69.5	86.9	68.3	83.9	67.2	81.4	11.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.7	91.8	72.2	63.6	88.4	71.4	62.5	85.0	70.8	33.6	87.2	32.5	84.3	31.7	81.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.7	21.5	19.2		DB	35.8	103.0	2.1	2.5	34.3	104.8	33.1	106.3	31.9	107.7	30.4	109.5
				WB	34.3	71.8	2.3	1.5	32.7	72.9	31.4	73.8	30.1	74.6	28.4	75.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	71.2	71.8	69.3	65.6	60.2	55.8	54.9	55.2	58.5	62.7	65.4	69.2	(6)
	DBStd	7.81	4.66	4.91	5.28	6.09	4.83	4.00	4.53	4.16	5.25	5.82	4.77	4.61	(7)
	HDD50	12	0	0	0	0	0	2	6	3	1	0	0	0	(8)
	HDD65	1531	3	2	12	65	167	279	313	304	209	116	51	10	(9)
	CDD50	4855	658	610	599	467	315	175	159	166	256	393	461	596	(10)
	CDD65	898	195	192	146	81	17	2	2	2	14	44	62	141	(11)
	CDH74	8436	1802	1687	1260	731	205	44	42	64	197	528	660	1216	(12)
	CDH80	3199	730	704	486	286	50	4	4	9	63	196	226	441	(13)
(14) Wind	WSAvg	8.4	10.6	10.0	8.8	7.4	6.7	6.6	6.6	7.0	7.5	8.9	10.0	10.3	(14)
(15) Precipitation	PrecAvg	10.5	0.3	0.2	0.3	0.7	1.3	1.9	1.8	1.8	1.0	0.5	0.5	0.3	(15)
	PrecMax	16.4	1.6	1.1	0.9	2.2	3.1	4.1	4.6	3.4	2.9	2.1	1.5	0.9	(16)
	PrecMin	4.6	0.0	0.0	0.0	0.1	0.1	0.2	0.5	0.4	0.2	0.0	0.0	0.0	(17)
	PrecStd	2.8	0.4	0.3	0.3	0.5	0.7	0.9	0.9	0.7	0.7	0.5	0.4	0.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.6	99.2	98.2	94.3	85.8	79.2	78.8	80.8	87.3	93.3	93.0	95.7	(19)
		MCWB	69.0	68.5	67.4	64.4	60.8	57.6	58.1	59.3	60.3	63.6	64.4	66.7	(20)
	2%	DB	92.7	93.4	90.8	87.9	80.3	73.7	73.6	74.0	79.2	86.1	86.4	89.7	(21)
		MCWB	66.4	67.5	66.4	63.3	59.6	56.9	56.9	57.3	60.2	61.8	62.4	65.6	(22)
	5%	DB	87.9	88.3	85.2	82.5	74.6	68.3	69.2	68.2	73.6	80.4	81.7	84.8	(23)
		MCWB	66.3	66.5	64.7	62.1	58.7	56.0	56.1	56.4	58.7	60.2	61.5	64.7	(24)
	10%	DB	83.4	83.8	80.7	77.3	69.6	64.0	64.2	64.0	68.4	75.0	77.0	80.5	(25)
		MCWB	64.9	65.8	64.4	61.2	58.4	55.3	54.8	55.6	57.3	59.6	60.9	63.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	72.1	70.1	67.6	63.9	61.2	60.7	61.9	64.8	66.5	67.8	69.5	(27)
		MCDB	90.5	89.3	90.3	84.8	76.4	70.9	72.0	74.3	80.2	86.7	83.1	87.9	(28)
	2%	WB	69.2	69.7	68.3	65.7	62.2	59.4	58.6	59.4	61.6	63.8	64.9	67.7	(29)
		MCDB	85.8	86.2	83.8	81.0	73.3	66.4	67.7	68.4	74.2	79.1	78.7	83.5	(30)
	5%	WB	67.6	68.2	66.9	64.1	60.8	58.2	57.2	57.7	59.6	62.0	63.4	66.4	(31)
		MCDB	82.3	83.0	80.3	76.6	69.7	63.9	65.2	64.9	69.8	75.1	75.9	79.9	(32)
	10%	WB	66.4	66.8	65.6	62.6	59.5	57.0	56.0	56.3	58.2	60.5	62.1	65.1	(33)
		MCDB	79.1	80.3	77.4	73.0	66.4	61.8	62.3	62.5	66.4	71.2	73.1	77.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	22.0	21.6	21.1	18.6	18.2	19.5	18.6	20.1	21.5	20.6	21.1	(35)
		MCDBR	29.9	29.4	28.7	29.1	28.1	27.2	27.9	27.9	29.1	31.0	29.9	29.0	(36)
		MCWBR	9.9	9.6	9.8	10.2	12.4	14.1	14.7	14.8	13.2	12.0	11.0	9.9	(37)
	5% WB	MCDBR	23.9	24.0	23.4	23.7	21.8	18.3	22.0	21.7	23.8	24.7	23.3	23.1	(38)
(39)		MCWBR	9.2	9.2	9.2	10.0	11.7	11.4	13.5	13.1	12.4	11.0	9.6	8.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.336	0.336	0.331	0.318	0.315	0.309	0.312	0.329	0.338	0.348	0.332	0.336	(40)
	taud		2.534	2.540	2.550	2.581	2.581	2.570	2.555	2.509	2.446	2.435	2.522	2.526	(41)
	Ebn at Noon		318	311	302	289	273	266	271	280	294	304	317	319	(42)
	Edh at Noon		35	34	31	27	24	23	24	29	34	37	35	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2637	2372	1926	1414	989	844	941	1176	1610	2092	2476	2662	(44)
(45)	RadStd		66	65	64	73	67	48	56	53	62	92	69	83	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	(47)

Nomenclature: See separate page